

Diagnostic Accuracy and Concordance of Self-Collected Cervical CDO1 and CELF4 Methylation Testing Compared With Physician Sampling for Endometrial Cancer Detection

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ABSTRACT

PURPOSE To address the growing incidence of endometrial cancer (EC), noninvasive detection methods are needed. Although self-sampling is proven for cervical cancer screening, its utility for EC detection via methylation analysis lacks robust evidence. This study aimed to validate this approach by evaluating the concordance and diagnostic accuracy of *CDO1*/*CELF4* methylation testing in self-collected samples compared with clinician-collected samples.

METHODS In this prospective cross-sectional study, paired self-collected (Evalyn Brush) and physician-collected cervical samples were obtained from women referred for hysteroscopy. *CDO1* and *CELF4* methylation levels were measured by qPCR. Method agreement was analyzed with Spearman correlation and Bland-Altman plots, and diagnostic performance was evaluated by the area under the ROC curve (AUC).

RESULTS Analysis of 142 participants revealed high agreement between self- and physician-collected samples, with significant correlations (*CDO1* $R = 0.79$, *CELF4* $R = 0.76$; $P < .05$) and Bland-Altman analysis confirming minimal bias ($P > .05$ for both genes). Self and physician sampling showed comparable efficacy for EC detection via *CDO1*/*CELF4* methylation, with AUCs of 0.930 and 0.896, respectively. Methylation levels (ΔCt) differed significantly ($P < .001$) between EC ($n = 33$) and non-EC ($n = 109$) groups, regardless of the collection method. The diagnostic performance remained consistently high across premenopausal and postmenopausal subgroups.

CONCLUSION Self-collected vaginal samples for *CDO1*/*CELF4* methylation analysis demonstrate high agreement and diagnostic accuracy equivalent to physician-collected samples for EC detection. This self-sampling strategy represents a reliable, noninvasive, and patient-centric approach, showing potential for integration into future EC diagnosis strategies to improve accessibility and early detection. Future multicenter studies are warranted to validate its effectiveness in broader populations.

ACCOMPANYING CONTENT

 Graphical Abstract
 Data Supplement

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INTRODUCTION

Endometrial cancer (EC) has become the most common gynecologic malignancy in high-income countries.¹ Survival declines sharply with stage, from 92% in stage I to 14% in stage IV.² Moreover, the increasing incidence of uterine cancer, together with its distinct epidemiologic patterns in women younger than 50 years, underscoring the importance of early detection and timely diagnosis.³

Current clinical evaluation for suspected EC includes pelvic examination, imaging—most commonly transvaginal

ultrasound (TVS)—and endometrial biopsy. Although these approaches are effective in many settings, they require specialist referral and involve invasive procedures. TVS is the gold standard for triaging women with abnormal uterine bleeding (AUB), but guideline-recommended endometrial thickness (ET) thresholds for histologic assessment in peri/postmenopausal women vary internationally, ranging from >3 mm in German and ESMO guidelines to >4 –5 mm in the United States and United Kingdom.⁴ Furthermore, although endometrial biopsy on the basis of Pipelle and Tao Brush is substantially accurate, less invasive, and costly than hysteroscopy, it was constrained by painful procedure

CONTEXT

Key Objective

Can self-collected cervical/vaginal samples analyzed using DNA methylation markers provide diagnostic accuracy for endometrial cancer (EC) comparable with physician-collected samples, thereby offering a reliable, noninvasive alternative within the diagnostic pathway for women undergoing evaluation for abnormal uterine bleeding?

Knowledge Generated

In this comparative study of 142 women undergoing hysteroscopy for clinical indications, self-collected samples demonstrated high diagnostic accuracy for EC using *CDO1* and *CELF4* DNA methylation markers. Diagnostic performance of self-sampling was comparable with physician-collected sampling, with consistently high discrimination across premenopausal and postmenopausal subgroups and robust findings in sensitivity analyses accounting for missing data.

Relevance

Self-collected DNA methylation testing represents a feasible, patient-acceptable triage approach within the diagnostic evaluation of symptomatic women, with potential to reduce reliance on invasive procedures without compromising diagnostic accuracy. This strategy may improve patient experience and optimize diagnostic workflows before hysteroscopy or other invasive assessments.

and inadequate samples, especially in nulliparous women.⁵ Consequently, there remains an unmet need for minimally invasive, accurate methods that can improve the diagnostic pathway for EC.

DNA methylation tests based on upper genital tract shed cells, which can be collected from cervical scrapings,⁶ had been proved as promising cancer biomarkers.^{7,8} A qPCR assay detecting cysteine dioxygenase type 1 (*CDO1*) and CUGBP Elav-like family member 4 (*CELF4*) gene methylation in cervical scrapings showed higher EC diagnostic accuracy than ET measurement. It demonstrated high sensitivity (87.5% postmenopausal bleeding [PMB], 85.7% premenopausal AUB) and specificity (95.9% PMB, 87.6% AUB), highlighting strong potential as a noninvasive diagnostic tool.^{9,10}

The successful integration of self-sampling for human papillomavirus (HPV)-DNA testing into cervical cancer screening guidelines has dramatically expanded screening coverage, particularly among women in remote areas or those with limited access to health care facilities¹¹⁻¹³; its application in the context of EC detection remains considerably less developed. Research on self-sampling methods, such as tampons and vaginal secretions, for the detection of EC is emerging, with a current emphasis on identifying methylation biomarkers.¹⁴⁻¹⁶

This prospective study evaluated the diagnostic accuracy and concordance of *CDO1/CELF4* methylation testing between self- and physician-collected cervical samples for EC detection. The primary end point was the ability of *CDO1/CELF4* methylation testing to discriminate EC from nonmalignant conditions, assessed by the area under the receiver operating characteristic (ROC) curve (AUC), with direct comparison between self-collected and physician-collected

cervical samples. Secondary objectives were to evaluate diagnostic performance stratified by menopausal status, compare methylation testing with conventional diagnostic modalities, and assess agreement between sampling methods. Exploratory analyses assessed associations between methylation levels and clinicopathologic features.

METHODS

Study Design and Participants

In this prospective cross-sectional study, adult women undergoing hysteroscopy were recruited from the Second Xiangya Hospital of Central South University, from March 2024 to February 2025. Women were prospectively enrolled if they were scheduled to undergo hysteroscopy for clinical indications, including AUB, abnormal TVS findings, or other clinical concerns requiring further endometrial evaluation. Accordingly, all participants were symptomatic or had abnormal clinical or imaging findings, and the study population represents women undergoing diagnostic evaluation rather than an asymptomatic screening population. Exclusion criteria included (1) current menstruation, pregnancy, or lactation; (2) recent hormonal therapy; (3) concurrent malignancies or active autoimmune diseases; (4) psychiatric disorders impairing consent/compliance; (5) intrauterine device use; (6) recent intrauterine procedures/trauma (<1 month); and (7) incomplete data or unavailable pathologic images. All participants provided informed consent per STROBE guidelines.¹⁷

Quality Control for Data Collection

To minimize selection bias and ensure transparent reporting, all eligible women referred for hysteroscopy during

the study period consecutively enrolled with prospective documentation of exclusions. Physician- and self-collected specimens underwent identical transport and laboratory procedures, with laboratory staff blinded to clinical information. Trained personnel systematically collected clinical data (age, symptoms, medical history, BMI) and diagnostic indicators (ET, serological markers, hysteroscopic and pathology findings) following a predefined protocol. Menopause was defined as ≥ 12 months of amenorrhea in women >40 years (excluding pregnancy/other causes).

Sample Collection

After informed consent, participants received a self-sampling kit with an Evalyn Brush (Rovers Medical Devices, Oss, Netherlands) and cell preservation solution (Cidabio Biotech, #Bio-CD301-1). With one foot elevated, they inserted the brush until the flanged neck secured at the vaginal introitus, rotated it five full turns, and detached the brush head into the preservation solution. Tubes were capped, stored at ambient temperature, returned to staff, and transported to the laboratory (Fig 1A). Physician-sampled cervical specimens were obtained using a CervexBrush (Rovers Medical Devices) rotated five times clockwise during clinic visits (Fig 1B). Self-samples used Cidabio preservation solution (#Bio-CD301-1); physician samples used PreservCyt medium (Hologic, #70098-002). Samples were stored at room temperature for 7 days before testing; residuals were stored at -80°C .

CD01 and CELF4 Methylation Test

Methylation detection was performed by staff blinded to clinical data. Genomic DNA was extracted from samples using the JH-DNA Isolation Kit (OriginPoly Bio-Tec, Beijing, China) and quantified via NanoDrop 2000c (Thermo Fisher Scientific, DE). 500 ng gDNA underwent bisulfite conversion with the JH-DNA Methylation-Lightning MagPrep kit (OriginPoly Bio-Tec). *CD01/CELF4* methylation levels were measured via CISENDO DNA Methylation Detection Kit (real-time PCR, OriginPoly Bio-Tec; NMPA Cert. No. 20243402610) with GAPDH as internal control on SLAN-96P PCR system (Shanghai Hongshi Medical Tech, China). High methylation was defined by ΔCt values ($\Delta\text{Ct}_{\text{CD01}} = \text{Ct}_{\text{CD01}} - \text{Ct}_{\text{GAPDH}}$; $\Delta\text{Ct}_{\text{CELF4}} = \text{Ct}_{\text{CELF4}} - \text{Ct}_{\text{GAPDH}}$). Assays were valid if $\text{Ct}_{\text{GAPDH}} < 32.2$ (per manufacturer's protocol). A positive result for the *CD01/CELF4* methylation assay was defined as a positive test for either hypermethylated *CD01* or *CELF4* gene, and the *CD01* & *CELF4* methylation assay was defined as a positive test for both hypermethylated *CD01* or the *CELF4* gene.

Transvaginal Ultrasound Detection

ET, echotexture abnormalities, the presence of hydrops, and blood flow signals were evaluated with the Hitachi Aloka Noblus TVS system (Hitachi Healthcare, Tokyo, Japan).

Serum Biomarkers

The levels of serum biomarkers, including carbohydrate antigen 125 (CA125, Carbohydrate Antigen 125 Quantitative Detection Kit, Tellgen, China), carbohydrate antigen 199 (CA199, Carbohydrate Antigen 199 Quantitative Detection Kit, Tellgen, China), carcinoembryonic antigen (CEA, Carcinoembryonic Antigen Assay Kit, Tellgen, China), and human epididymis protein 4 (HE4, Human Epididymis Protein 4 Quantitative Detection Kit, Tellgen, China), were evaluated through fully automated high-throughput immunoassay system (Tellgen, China) according to the prescribed protocol.

Vaginal Secretion Examination

Microbial density, microbial diversity, pH, Nugent score, and bacterial vaginosis (BV) status of patient vagina microenvironment were measured by vaginal microecology evaluation system (Bioperfectus, Jiangsu, China).

Pathology

Pathologic diagnosis was established by two gynecologic pathologists blinded to sampling method and methylation results, with discrepancies resolved by a senior pathologist. Endometrial pathology was classified according to the 2020 WHO Classification of Female Genital Tumors (5th edition). The system categorizes lesions into premalignant lesions and malignant carcinomas. The premalignant category comprises endometrial hyperplasia (EH) without atypia (a benign proliferation with minimal cancer risk) and endometrial atypical hyperplasia/endometrioid intraepithelial neoplasia (EAH/EIN). For patients with EC, the histologic type, grade, and stage were determined using the International Federation of Gynecology and Obstetrics (FIGO) 2023 staging system.¹⁸

Statistical Analysis

In this study, the positive findings of TVS evaluation were defined as ET ≥ 11 mm (premenopausal) and ET ≥ 5 mm (postmenopausal). BMI ≥ 25 kg/m², CA125 ≥ 35 U/mL, CEA ≥ 35 ng/mL, CA19-9 ≥ 35 U/mL, and HE4 ≥ 70 pmol/L for the premenopausal/ ≥ 14 pmol/L are defined as abnormal. Statistical analysis was performed using R version 4.4.3. The sample size was estimated based on the primary end point, the area under the ROC curve (AUC), using a precision-based approach. Assuming an anticipated AUC of 0.85, a two-sided α of 0.05, and a desired half-width of 0.05 for the 95% CI, the estimated minimum total sample size was 102. In the main text, categorical data are presented as counts and percentages, and group comparisons were conducted using the chi-squared test. Continuous data, which did not strictly conform to a normal distribution in this study, are presented as median and interquartile range, and comparisons were performed using the Wilcoxon test. Univariable logistic regression were used to analyze the risk factors associated with

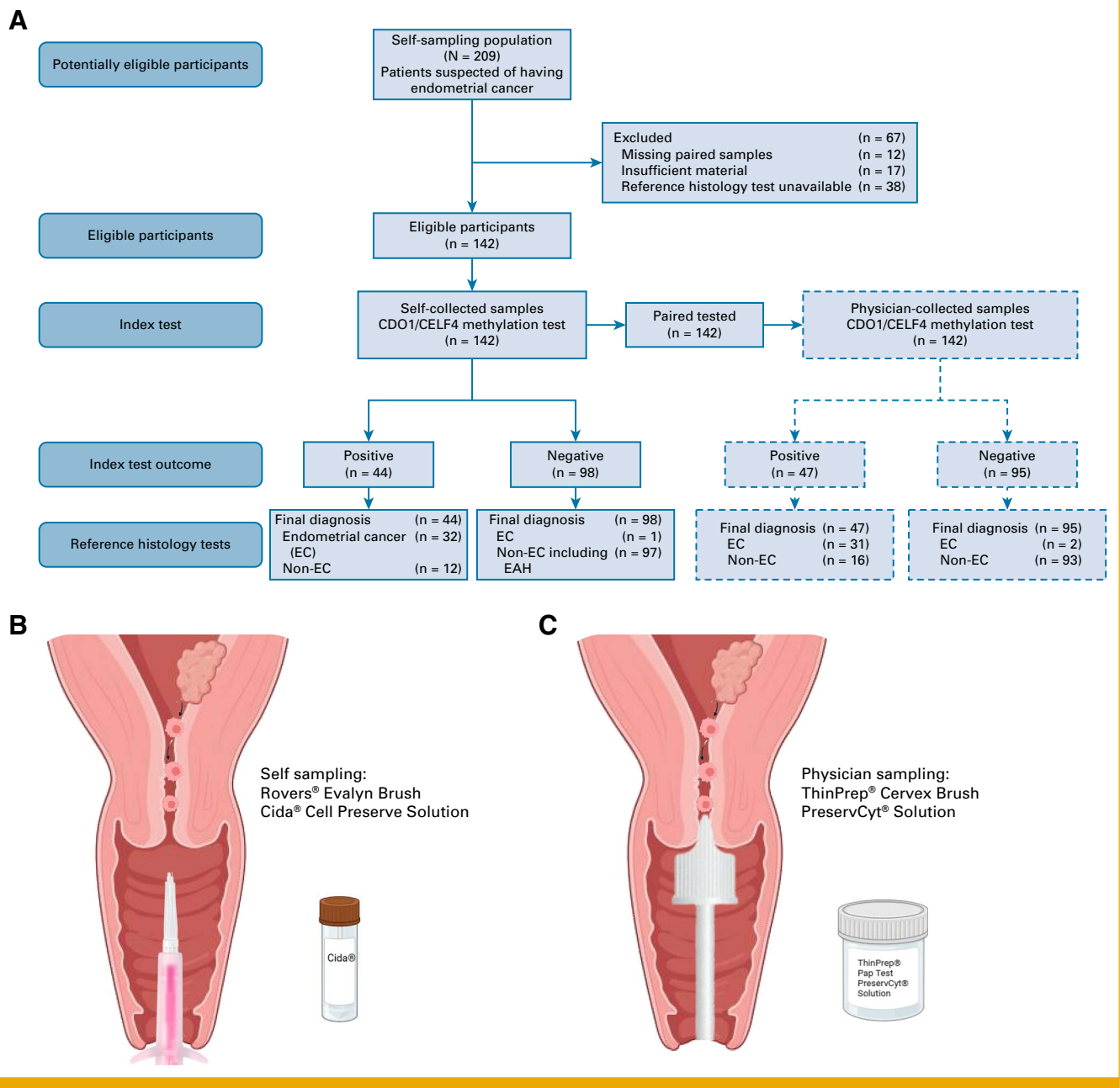


FIG 1. Research program and sampling methods. (A) Research program; (B) self-sampling method with Evalyn Brush and Cida preservation solution; (C) physician-sampling method with Cervex Brush and PreservCyt solution.

EC, calculating odds ratios (ORs) along with their corresponding 95% CIs. Logistic regression analyses were conducted for exploratory association assessment only and were not intended for risk prediction or model development. Spearman correlation assessed monotonic relationships between sampling methods. Bland-Altman analysis evaluated intermethod agreement. ROC curves were generated to assess diagnostic performance, with AUC, sensitivity, and specificity calculated. The DeLong test compared AUCs between diagnostic strategies. A two-tailed P value $<.05$ indicated statistical significance. Complete-case analysis was used as

methylation data were complete. Baseline characteristics of excluded ($n = 67$) versus included ($n = 142$) participants were compared with assess selection bias. Sensitivity analyses using multiple imputation by chained equations (MICE) evaluated the impact of missing covariates on risk factors and diagnostic performance, with results pooled using Rubin's rules. Given the exploratory nature of secondary and subgroup analyses, P values were not adjusted for multiple testing. All primary analyses were prespecified, and key findings were supported by large effect sizes and consistent results across sampling methods and menopausal strata.

Ethics Statement

This study received approval from the Institutional Review Board (IRB) of the Ethics Committee at Xiangya Second Hospital, Hunan, China (IRB No.: LYEC2025-0003).

RESULTS

Baseline Characteristics of Participants

The final cohort included 142 participants from 209 hysteroscopy-indicated women after excluding 67 with unpaired samples, insufficient cells, or incomplete data (Fig 1A); all included participants had valid *CDO1/CELF4* methylation results from both physician- and self-collected samples (Figs 1B and 1C). Among patients with available data, no statistically significant differences were observed between excluded and included participants in age, BMI, menopausal status, or comorbidities (Table S1). However, it is important to note that the proportion of missing data for certain baseline characteristics was notably higher in the excluded group than in the included group. The non-EC group ($n = 109$) consisted of 35% EH, 29% endometrial metaplasia (EM), 26% endometrial polyps (EP), and 10% EAH. The EC group ($n = 33$) was predominantly composed of endometrioid carcinoma (88%) with the remainder being serous carcinoma (12%). The majority of EC cases were stage I (74%), followed by stage II (26%; Data Supplement, Table S2). The EC group was considerably older than the non-EC group (median age: 57.00 v 48.50 years; $P < .001$), with a correspondingly higher proportion of patients older than 50 years (76% v 37%; Table 1 and Data Supplement, Table S3). Consistent with the age profile, postmenopausal status was also more frequent in the EC group (24/59 v 9/81), whereas no significant difference was observed in BMI between the groups ($P = .6$, Table 1). Significantly higher methylation levels (lower ΔCt values) for both *CDO1* and *CELF4* were observed in patients with EC compared with controls regardless of sampling method, a distinction that remained highly significant (all $P < .001$) regardless of menopausal status (Table 1).

Demographically, patients with EC had a higher average menarche age (13.91 v 13.20 years, $P = .026$), and no patients with EC had thyroid disorders history (0% v 22%) compared with non-EC patients; symptoms including AUB ($P = .087$), abnormal discharge ($P = .079$), irregular menstruation ($P = .3$), and abdominal discomfort ($P = .6$) showed no significant between-group differences, nor did characteristics such as rural/urban origin, BMI, gravidity, parity, diabetes history, and hypertension history (all $P > .05$; Data Supplement, Table S3). No systematic differences in pathologic or demographic characteristics were observed across menopausal status groups (Data Supplement, Tables S2 and S3).

Risk Factors for EC

Stratified analysis of standard diagnostic tests revealed significant performance disparities based on menopausal

status. As shown in the Data Supplement (Table S4), ET differed significantly between the EC and non-EC groups in premenopausal women ($P = .023$) but failed to distinguish EC in the postmenopausal cohort ($P = .42$). Similarly, serum biomarkers (CA125, CEA, and HE4) and vaginal microecology indicators (including Nugent score and BV) showed no significant diagnostic value across subgroups ($P > .05$), with the exception of a significantly higher vaginal pH observed in the overall EC group ($P = .03$) and premenopausal subgroup ($P = .011$).

In the univariable logistic regression analysis (Data Supplement, Table S5), age (OR, 1.088; 95% CI, 1.044 to 1.14; $P < .001$) and menopausal status (OR, 5.486; 95% CI, 2.38 to 13.65; $P < .001$) were identified as significant clinical risk factors. Conversely, BMI, diabetes, and hypertension were not significantly associated with EC in this cohort ($P > .05$). Methylation markers demonstrated exceptional predictive power for EC, far surpassing clinical characteristics (Data Supplement, Table S5). Higher ΔCt value for methylation tests acted as a protective effect. The combined *CDO1* and *CELF4* methylation test demonstrated consistently high ORs across different sampling methods, highlighting its robust independent predictive value for malignancy. Multiple imputation analysis yielded risk factor estimates consistent with complete-case analysis (eg, age OR 1.087 v 1.088; menopausal status OR 5.545 v 5.486; Data Supplement, Table S6).

DNA Methylation Biomarker Level Distribution

Given the known clinical differences between premenopausal and postmenopausal women, diagnostic performance was evaluated both in the overall cohort and stratified by menopausal status.

Both sampling methods consistently demonstrated that *CDO1* and *CELF4* methylation levels were significantly elevated (indicated by lower ΔCt values) in the EC group compared with all non-EC benign conditions, including hyperplasia and polyps ($P < .0001$; Figs 2A, 2B, 2C and 2D). Importantly, this robust discriminatory power persisted after stratifying by menopausal status (Table 1, Data Supplement, Table S4). Furthermore, methylation intensity correlated with disease progression; physician-collected samples showed significantly lower ΔCt values in stage II compared with stage I EC ($P < .05$; Figs 2E and 2F), a trend also mirrored in self-collected samples (Figs 2G and 2H).

Concordance and Agreement of Different Sampling Methods

Both methylated *CDO1* and *CELF4* showed strong concordance between sampling methods in Spearman correlation analysis (both $P < .001$). Methylated *CDO1* exhibited a slightly stronger correlation ($R = 0.79$) than *CELF4* ($R = 0.76$), as shown in Figure 3A and 3B.

TABLE 1. Characteristics of Participants

Characteristic	All Participants				Premenopausal				Postmenopausal			
	N	Non-EC, n = 109 ^a	EC, n = 33 ^a	P	N	Non-EC, n = 72 ^a	EC, n = 9 ^a	P	N	Non-EC, n = 35 ^a	EC, n = 24 ^a	P
Age group	141			<.001 ^b	81			.8 ^b	59			.13 ^b
<30		7 (6%)	—			6 (8%)	—			—	—	
30–50		61 (56%)	8 (24%)			54 (75%)	7 (78%)			7 (20%)	1 (4%)	
>50		40 (37%)	25 (76%)			12 (17%)	2 (22%)			28 (80%)	23 (96%)	
Missing		1	0									
BMI group	132			.6 ^b	78			.5 ^b	54			.7 ^b
<18.5		2 (2%)	1 (3%)			2 (3%)	—			—	1 (5%)	
18.5–24.9		64 (63%)	20 (65%)			40 (58%)	5 (56%)			24 (75%)	15 (68%)	
25–29.9		29 (29%)	7 (23%)			22 (32%)	2 (22%)			7 (22%)	5 (23%)	
>30		6 (6%)	3 (10%)			5 (7%)	2 (22%)			1 (3%)	1 (5%)	
Missing		8	2			3	0			3	2	
CD01 (self-sampled)	142	16.90 (14.40, 18.07)	6.99 (5.06, 8.04)	<.001 ^c	81	16.60 (14.38, 18.10)	5.53 (5.06, 7.04)	<.001 ^c	59	17.49 (14.11, 18.06)	7.03 (5.16, 8.77)	<.001 ^c
CELFA (self-sampled)	142	16.46 (13.59, 17.93)	7.12 (4.98, 9.74)	<.001 ^c	81	16.30 (14.24, 18.23)	6.68 (4.90, 8.42)	<.001 ^c	59	16.82 (12.28, 17.83)	7.54 (5.50, 10.11)	<.001 ^c
CD01 (physician-sampled)	142	16.17 (11.77, 17.73)	5.67 (4.08, 8.12)	<.001 ^c	81	16.16 (12.57, 17.76)	5.58 (4.45, 7.29)	<.001 ^c	59	16.33 (10.85, 17.80)	5.76 (3.92, 8.64)	<.001 ^c
CELFA (physician-sampled)	142	15.78 (11.37, 17.67)	7.12 (4.04, 9.36)	<.001 ^c	81	15.73 (11.53, 17.69)	5.59 (4.04, 8.47)	<.001 ^c	59	15.78 (11.19, 17.67)	7.17 (3.98, 9.59)	<.001 ^c

NOTE. Data on menopausal status were missing for two analyzed participants.

^aNo. (%); median (Q1, Q3).

^bFisher exact test.

^cWilcoxon rank sum test.

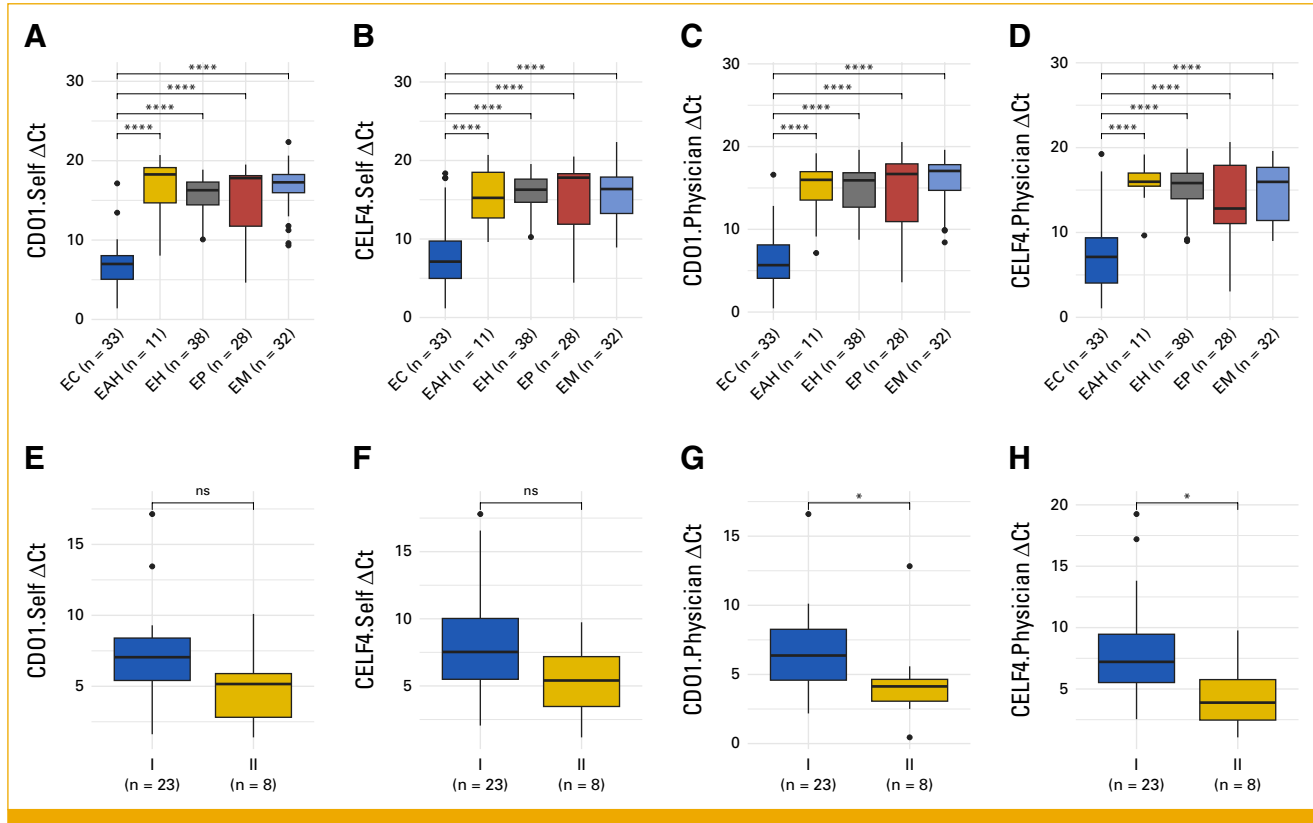


FIG 2. Distribution of *CDO1* and *CELF4* ΔC_t in different pathological groups. *CDO1* (A,C) and *CELF4* (B,D) ΔC_t distribution in EC, EH, EAH, EP, and EM for different sampling methods; *CDO1* (E,G) and *CELF4* (F,H) ΔC_t distribution in different EC stage I and II for different sampling methods. All y axis represents ΔC_t value of gene methylation test. * $P < .05$, **** $P < .0001$, ns $P > .05$. EAH, endometrial atypical hyperplasia; EC, endometrial cancer; EH, endometrial hyperplasia; EM, endometrial metaplasia; EP, endometrial polyps.

The self-sampling and physician-sampling methods showed comparable outcomes for detecting methylated *CDO1* and *CELF4*, with no significant systematic bias found in paired t-tests (*CDO1*: $P = .3114$; *CELF4*: $P = .1756$). Bland-Altman analysis revealed acceptable agreement between the methods, with a slightly tighter concordance for *CDO1* (mean bias: 0.49; limits of agreement [LOA], -4.90 to 5.89) than for *CELF4* (mean bias: 0.89; LOA, -6.34 to 8.12) in Figures 3C and 3D. The methylated *CDO1* gene shows marginally better consistency compared with *CELF4*.

Diagnosis Performance of DNA Methylation Test for Different Sampling Methods

Self-collected DNA methylation testing demonstrated diagnostic accuracy comparable with physician-collected sampling. In the overall cohort, self-sampled *CDO1* achieved an AUC of 0.967 (95% CI, 0.931 to 1.000), outperforming *CELF4* (AUC, 0.882; 95% CI, 0.801 to 0.963; Fig 4, Table 2).

Performance remained stable across menopausal strata. In premenopausal women, self-sampled *CDO1* showed perfect discrimination (AUC, 1.000), whereas *CELF4* achieved an AUC of 0.904 (95% CI, 0.759 to 1.000; Data Supplement, Fig S1). In postmenopausal women, high accuracy was maintained for

both markers (*CDO1*: AUC, 0.944; 95% CI, 0.884 to 1.000; *CELF4*: AUC, 0.864; 95% CI, 0.756 to 0.971; Data Supplement, Fig S2). Corresponding physician-collected samples yielded comparable AUCs, supporting the reliability of self-sampling (Table 2).

In self-collected samples, the inclusive *CDO1*(+)/*CELF4*(+) strategy demonstrated consistently high discrimination (overall AUC, 0.930; 95% CI, 0.888 to 0.972), with similar performance in premenopausal (AUC, 0.938; 95% CI, 0.899 to 0.976) and postmenopausal women (AUC, 0.936; 95% CI, 0.874 to 0.999). The stringent *CDO1*(+)&*CELF4*(+) strategy showed lower discrimination (AUC, 0.845) but higher specificity (96.3% overall; 95.8% postmenopausal). No meaningful differences were observed compared with physician-collected combined strategies (Table 2; Data Supplement, Table S7).

However, clinical assessment using ET and abnormalities defined by classic cutoffs showed limited diagnostic value (AUCs < 0.6). Stratified analysis revealed that although ET offered moderate diagnostic value in premenopausal women (AUC = 0.778), it failed to effectively discriminate EC in the postmenopausal subgroup (AUC = 0.578). A sensitivity analysis using imputed values for ET yielded similar ROC

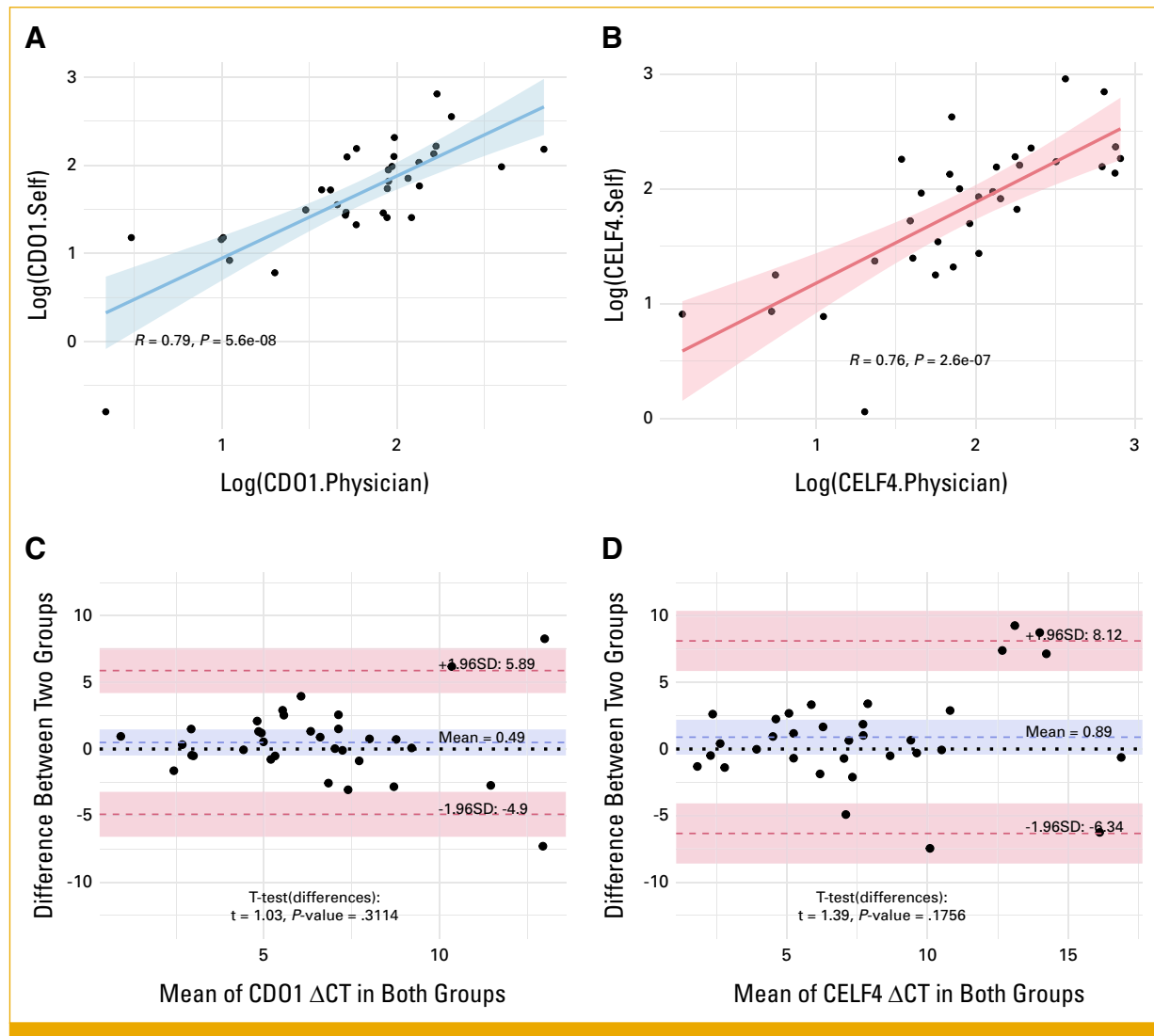


FIG 3. Consistency of self-sampling and physician-sampling *CDO1* and *CELF4* Δ CT values. (A, B) Correlation based on the log transformed Δ CT between two sampling groups. (C, D) Bland-Altman plots comparing both two sampling groups.

results, whereas methylation-based index tests showed identical performance because of complete data (Data Supplement, Table S8).

DISCUSSION

The global incidence of EC has been rising rapidly, now surpassing cervical cancer as the most common gynecologic malignancy in high-income countries.¹⁹ Developing less invasive, patient-friendly, and accurate detection methods is essential to improve early diagnosis and surveillance. This study evaluated the diagnostic agreement and accuracy of self-collected versus physician-collected cervical scrapings for *CDO1/CELF4* methylation-based EC detection. Both sampling methods demonstrated significant *CDO1/CELF4* methylation differences between EC and non-EC groups, which is consistent with menopausal strata analysis, confirming the robustness of these markers (Table 1). Compared

with other detection methods, methylation detection has a unique independent predictive value for EC (Data Supplement, Table S5). Bland-Altman analysis indicated minimal bias between sampling methods (Fig 3). The dual-methylation (*CDO1/CELF4*) test achieved markedly higher accuracy than ET measurement (Fig 4, Table 2) regardless of menopausal status. These findings support the reliability of self-collected samples as a viable approach for early EC detection.

Serum biomarkers such as CA125, CA199, CEA, and HE4 are widely used in gynecologic oncology but exhibit low diagnostic accuracy for EC. Even in combination (HE4+CA125+CA199, AUC 0.79), their performance remains inadequate for clinical early diagnosis.^{20,21} Consistent with prior studies, our findings confirm that these biomarkers lack sufficient sensitivity and specificity for EC detection. Routine lower genital tract noninvasive tests (HPV, liquid-based cytology, or

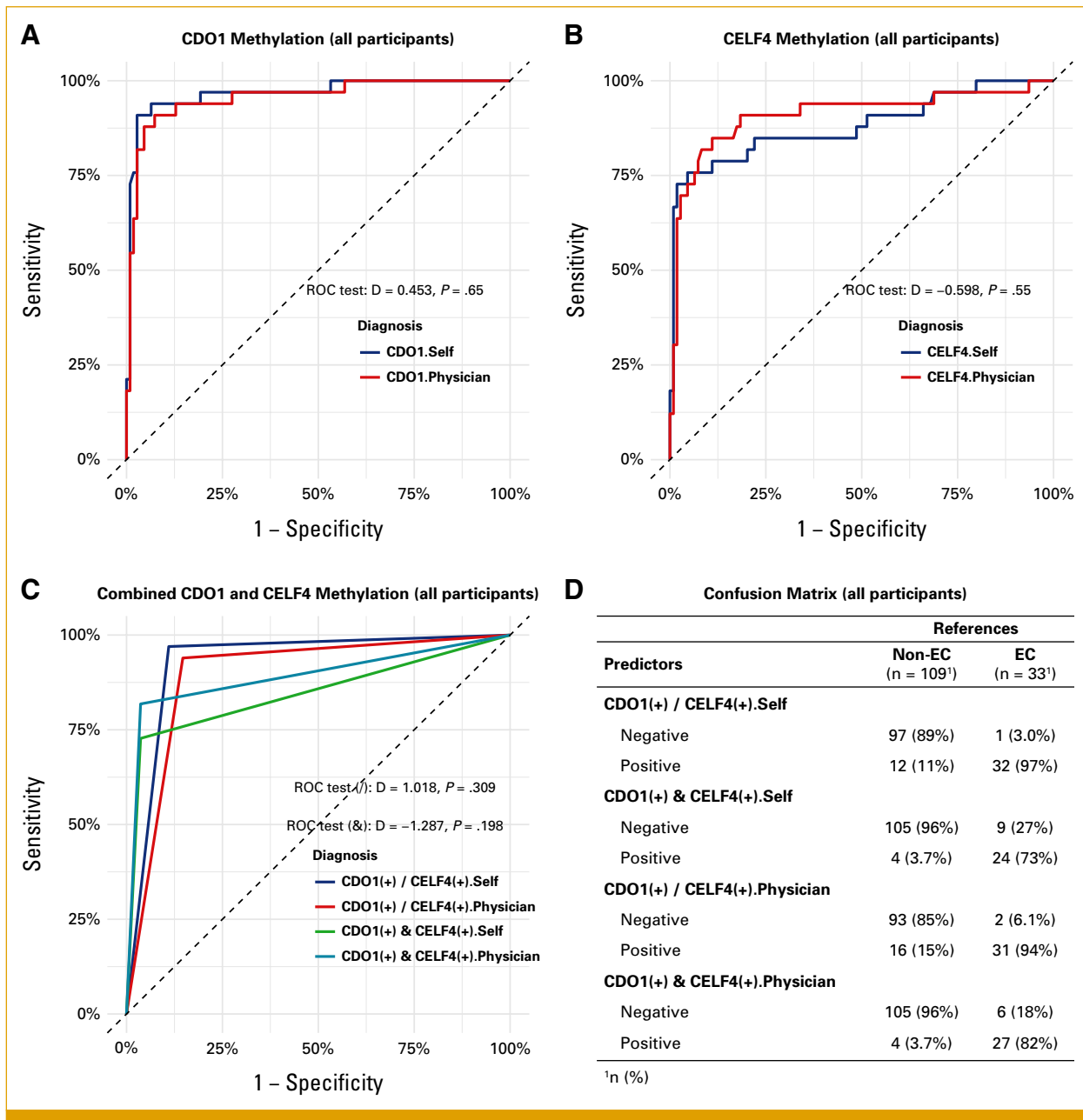


FIG 4. ROCs for different methods. ROC of *CDO1* (A), *CELF4* (B) methylation, and combined tests (C) both physician-sampled and self-sampled samples; (D) Confusion matrix for combined tests. ROCs, receiver operating curves.

microecology) are also unsuitable for identifying early EC (Data Supplement, Table S4). Interestingly, we observed elevated vaginal pH among patients with EC (Data Supplement, Table S4), suggesting a high-pH microenvironment that may promote microbial dysbiosis, aligning with previous observations,²² but there was nonsignificant difference in menopausal status strata analysis. Further microbiome-based studies are needed to validate these findings.

Self-sampling has gained attention as a minimally invasive, patient-centered approach for specimen collection.²³ In cervical cancer screening, self-sampling HPV test

achieves comparable detection rates for CIN2+ and CIN3+ (74%-86% and 75%-86%, respectively), with markedly higher acceptance compared with physician sampling.²⁴ Its advantages—privacy, comfort, convenience, and reduced embarrassment—drive high willingness to adopt and repeat self-sampling (97% and 87%, respectively).²⁵ Building on this success, an increasing number of studies have explored self-sampling methods for EC detection, primarily employing gene methylation assays,^{15,16,26-28} while others have investigated cytology,²⁹ gene mutations,³⁰ and proteomics.³¹ Women's acceptance of self-sampling for EC testing is consistently high.³² Various sample types have been tested, including tampons,^{15,26} urine,^{27-29,31} and vaginal

TABLE 2. Diagnostic Performance of Methylation Tests for EC

Method	All			Premenopausal			Postmenopausal		
	AUC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
ET abnormality	0.577 (0.48 to 0.674)	0.724 (0.552 to 0.862)	0.430 (0.333 to 0.538)	0.675 (0.549 to 0.8)	0.889 (0.667 to 1.000)	0.460 (0.349 to 0.571)	0.508 (0.37 to 0.647)	0.650 (0.450 to 0.850)	0.367 (0.200 to 0.533)
<i>CDO1</i> .Self	0.967 (0.931 to 1)	0.939 (0.848 to 1.000)	0.963 (0.899 to 0.991)	1 (1 to 1)	1.000 (1.000 to 1.000)	1.000 (1.000 to 1.000)	0.944 (0.884 to 1)	0.917 (0.833 to 1.000)	0.943 (0.829 to 1.000)
<i>CDO1</i> .Physician	0.956 (0.915 to 0.996)	0.909 (0.818 to 1.000)	0.936 (0.862 to 0.982)	0.989 (0.972 to 1)	1.000 (1.000 to 1.000)	0.972 (0.931 to 1.000)	0.935 (0.866 to 1)	0.875 (0.750 to 1.000)	0.943 (0.829 to 1.000)
<i>CELFB4</i> .Self	0.882 (0.801 to 0.963)	0.818 (0.697 to 0.909)	0.908 (0.771 to 0.991)	0.904 (0.759 to 1)	0.889 (0.667 to 1.000)	1.000 (0.736 to 1.000)	0.864 (0.756 to 0.971)	0.792 (0.625 to 0.958)	0.914 (0.771 to 1.000)
<i>CELFB4</i> .Physician	0.909 (0.837 to 0.981)	0.879 (0.758 to 0.970)	0.890 (0.789 to 0.963)	0.887 (0.686 to 1)	0.889 (0.667 to 1.000)	0.986 (0.931 to 1.000)	0.91 (0.829 to 0.99)	0.875 (0.750 to 1.000)	0.886 (0.771 to 0.971)
<i>CDO1</i> (+)/ <i>CELF4</i> (+).Self	0.93 (0.888 to 0.972)	0.970 (0.909 to 1.000)	0.890 (0.826 to 0.945)	0.938 (0.899 to 0.976)	1.000 (1.000 to 1.000)	0.875 (0.792 to 0.944)	0.936 (0.874 to 0.999)	0.958 (0.875 to 1.000)	0.914 (0.829 to 1.000)
<i>CDO1</i> (+)/ <i>CELF4</i> (+).Physician	0.896 (0.843 to 0.949)	0.939 (0.848 to 1.000)	0.853 (0.789 to 0.917)	0.924 (0.882 to 0.965)	1.000 (1.000 to 1.000)	0.847 (0.764 to 0.931)	0.887 (0.805 to 0.968)	0.917 (0.792 to 1.000)	0.857 (0.743 to 0.971)
<i>CDO1</i> (+) & <i>CELF4</i> (+). Self	0.845 (0.766 to 0.924)	0.727 (0.576 to 0.879)	0.963 (0.927 to 0.991)	0.875 (0.73 to 1)	0.778 (0.444 to 1.000)	0.972 (0.931 to 1.000)	0.826 (0.725 to 0.926)	0.708 (0.500 to 0.875)	0.943 (0.857 to 1.000)
<i>CDO1</i> (+) & <i>CELF4</i> (+).Physician	0.891 (0.822 to 0.96)	0.818 (0.667 to 0.939)	0.963 (0.927 to 0.991)	0.924 (0.812 to 1)	0.889 (0.667 to 1.000)	0.958 (0.903 to 1.000)	0.882 (0.794 to 0.969)	0.792 (0.625 to 0.958)	0.971 (0.914 to 1.000)

Abbreviations: EC, endometrial cancer; ET, endometrial thickness.

secretions.^{16,28,29} Despite encouraging results, most studies remain limited sample size and exploratory. Collectively, these findings confirm the feasibility of methylation-based self-sampling for EC.

To translate self-sampling into real-world EC detection, we highlight the clinical applicability of established methylation tests such as the *CDO1/CELF4* assay, which is already approved for clinical use in China. In our data, *CDO1/CELF4* methylation showed strong correlation ($R > 0.75$) and high agreement (both $P > .05$) between self- and physician-collected samples (Fig 3), suggesting stable sample quality and consistent diagnostic performance (Table 2, Fig 4). Thus, a self-sampling workflow appears feasible for large-scale implementation.

In clinical practice, methylation assays can complement TVS to minimize missed diagnoses and overtreatment among women with unexplained AUB.³³ Although adjusting diagnostic thresholds improved TVS performance, methylation testing still showed superior accuracy, regardless of the menopausal status (Table 2). Because abnormal ET is common in benign conditions, TVS alone offers high sensitivity but limited specificity,³⁴ especially in premenopausal women (Table 2). Incorporating additional imaging parameters (echogenicity, vascularity) can improve discrimination,³⁵ while methylation testing serves as a

valuable triage tool for patients with abnormal TVS results.³³

Taken together, our findings support a feasible, self-sampling-based diagnostic strategy for EC, enabling at-home sample collection, mailing, and remote result review.

Nevertheless, some limitations must be acknowledged. The sample size was modest, and follow-up data were unavailable to assess long-term predictive performance. Additionally, diagnostic efficacy for EAH was not evaluated because of limited cases. As the study was conducted among women undergoing hysteroscopy for clinical indications, findings cannot be directly extrapolated to asymptomatic screening. Future multicenter prospective studies with larger cohorts and extended follow-up are warranted to validate the diagnostic value of methylation testing in EC.

In conclusion, the study establishes that *CDO1/CELF4* methylation analysis in self-collected cervical samples exhibits high diagnostic efficacy for EC, demonstrating excellent agreement with results from physician-collected samples and thereby supporting the reliability of self-sampling for standalone methylation testing. These findings provide initial evidence for the utility of self-sampling, suggesting its potential value for incorporation into future EC early detection strategy.

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DATA SHARING STATEMENT

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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